

DAYANANDA SAGAR COLLEGE OF ENGINEERING
(An Autonomous Institute affiliated to VTU, Belagavi - 590018)
Accredited by National Assessment & Accreditation Council (NAAC) with 'A' grade
Shavige Malleshwara Hills, Kumaraswamy Layout
Bengaluru-560111



Mini Project Report on

“Design and Implementation of a Multi-Factor Smart Security System with Cybersecurity Integration”

Submitted in partial fulfillment for the award of degree of

Bachelor of Engineering
in
Electrical and Electronics Engineering

Submitted by

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2024-2025

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING



CERTIFICATE

Certified that the mini project report entitled “**Design and Implementation of a Multi-Factor Smart Security System with Cybersecurity Integration**” carried out by **VARSHA H, N HEMANTH, NANDAN A P, PRADYUMNA G NADIGER** bearing **1DS23EE118, 1DS24EE406, 1DS24EE407, 1DS24EE408** are bona fide students of **DAYANANDA SAGAR COLLEGE OF ENGINEERING**, an autonomous institution affiliated to VTU, Belagavi in partial fulfillment for the award of Degree of **Bachelor of Engineering in Electrical and Electronics Engineering** during the year **2025-2026**. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The mini project report has been approved as it satisfies the academic requirements in respect of work prescribed for the said Degree.

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DECLARATION

We, Varsha H(1DS23EE118), N Hemanth(1DS24EE406), Nandan A P(1DS24EE407) and Pradyumna G Nadiger (1DS24EE408), respectively, hereby declare that the mini project work entitled “Design and Implementation of a Multi-Factor Smart Security System with Cybersecurity Integration” has been independently done by us under the guidance of ‘Dr. RahulKumar J, Assistant Professor, EEE department and submitted in partial fulfillment of the requirement for the award of the degree of Bachelor of Engineering in Electrical & Electronics Engineering, at Dayananda Sagar College of Engineering, an autonomous institution affiliated to VTU, Belagavi during the academic year 2024-2025.

We further declare that we have not submitted this report either in part or in full to any other university for the award of any degree.

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CO-PO Mapping

COURSE OUTCOMES

CO1: Identify a research problem by doing literature survey with supervisor's advice.

CO2: Apply the fundamental knowledge of mathematics, science and engineering principles in design of solutions of system components.

CO3: Select and apply a suitable engineering/IT tool in modeling/data interpretation /analytical studies, conduct experiments leading to a logical solution.

CO4: Design a system/ system component/process, build it and test its functioning as a solution to a complex engineering problem with sustainable practice.

CO5: Communicate effectively to a diverse audience about the outcome and societal impact; develop technical reports and publication.

CO-PO mapping:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO 3
CO 1		2											2	2	1
CO 2	1												2	2	1
CO 3	2				2								2	2	1
CO 4		3	3	3	3	3	3						2	2	1
CO 5						3		3		3		3	2	2	1

	PO1		PO3	PO4	PO5	PO6	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1		2									2	2	1
CO2	1										2	2	1
CO3	2				2						2	2	1
CO4		3	3	3	3	3					2	2	1
CO5						3	3		3	3	2	2	1

ABSTRACT:

The main focus of this project is to design and implement a Multi-Factor Smart Security System integrated with cyber security features to enhance protection against unauthorized access. Traditional security systems relying on a single authentication method are vulnerable to breaches, making it essential to adopt a multi-layered approach. This project combines multiple authentication factors such as password-based entry, biometric verification, and alert messages to ensure a higher level of security.

The system is developed using embedded hardware components and software integration to provide real-time monitoring and secure access control. Sensors, microcontrollers, and communication modules are utilized to detect and respond to intrusion attempts. In addition to physical security, the system incorporates cyber security measures such as data encryption, secure communication protocols, and protection against unauthorized network access. These features help safeguard sensitive user data and prevent Cyber threats

The implementation includes user authentication, alert generation, and remote access capabilities through a mobile or web-based interface. The system is designed to be reliable, scalable, and efficient for applications in homes, offices, and restricted areas. Experimental results demonstrate improved security performance compared to conventional systems by reducing the risk of unauthorized entry and cyber attacks.

This project highlights the importance of integrating physical security with cybersecurity to build a robust and intelligent security solution for modern application

Chapter – 1

Introduction

1.1. Overview

With the rapid growth of digital and connected systems, security has become a major concern in both physical and cyber domains. Traditional single-factor methods like passwords or keys are vulnerable to theft and hacking. Multi-factor authentication (MFA), which combines multiple verification methods such as passwords and biometrics, provides improved security. Additionally, the rise of smart and IOT-based systems has increased the need for cybersecurity measures like encryption and secure communication. This project focuses on developing a smart security system that integrates multi-factor authentication with cybersecurity to ensure reliable and efficient protection.

1.2. Problem statement

In recent years, the rise in security breaches across both physical and digital environments has exposed the weaknesses of traditional security systems. Most existing systems rely on single-factor authentication methods such as passwords, PINs, or keys, which are highly vulnerable to theft, duplication, and unauthorized access. Even systems that use biometric or alert message based authentication often function independently without proper integration, leading to potential security gaps and reduced reliability.

Moreover, the increasing use of smart and IOT-based security systems has introduced new cybersecurity challenges, including hacking, data breaches, and unauthorized remote access. Many current solutions lack essential cybersecurity features such as encryption and secure communication protocols, putting sensitive user data at risk. Therefore, there is a need for an integrated system that combines multi-factor authentication with strong cybersecurity measures to provide a reliable and comprehensive security solution.

1.3. Applications

The Multi-Factor Smart Security System with cybersecurity integration can be applied in various real-world scenarios where both physical safety and data protection are critical.

1.3.1. In residential security

The system enhances home safety by allowing only authorized individuals to enter through multiple authentication methods such as password or biometric verification. It can also send real-time alerts to homeowners in case of suspicious activity, ensuring continuous monitoring even when they are away.

1.3.2. In offices and workplaces

The system provides secure access control for employees and visitors, protecting sensitive areas like cabins, record rooms, and IT sections. It helps prevent unauthorized access and ensures that only verified personnel can enter restricted zones, thereby maintaining confidentiality and security.

1.3.3. In banking and financial institutions

The system can be used to secure vaults, lockers, and transaction areas. Multi-factor authentication significantly reduces the risk of fraud and unauthorized access, while cybersecurity measures protect financial data from cyber-attacks.

1.3.4. In educational institutions

The system can restrict access to laboratories, server rooms, and administrative offices. It ensures that only authorized staff and students can enter specific areas, improving overall campus security.

1.3.5. In data centres and server rooms

Where sensitive information is stored, the system provides both physical and cyber protection. It prevents unauthorized entry and secures communication networks using encryption, reducing the risk of data breaches.

Additionally, in smart buildings and IOT-based environments, the system can be integrated with other smart devices to enable automated and intelligent security management. This includes remote monitoring, access control via mobile applications, and real-time notifications, making the system highly efficient and user-friendly.

1.4. Scope of the project

The scope of this project focuses on the design and implementation of a Multi-Factor Smart Security System integrated with cybersecurity features to provide enhanced protection against

unauthorized access. The system is intended to combine multiple authentication methods such as password, biometric verification, to ensure a higher level of security compared to traditional systems. It includes the use of embedded hardware components like sensors and microcontrollers for access control, along with software integration for user authentication and system management. The project also covers the implementation of basic cybersecurity measures such as data encryption and secure communication to protect sensitive information from potential cyber threats. Additionally, the system provides features like real-time monitoring, alert generation, and limited remote access through a mobile or web interface. The proposed solution is suitable for applications in homes, offices, and small-scale secure environments. However, large-scale deployment, advanced AI-based threat detection, and high-end industrial security integrations are beyond the scope of this project.

1.5. Motivation

The increasing number of security breaches, both physical and cyber related, has highlighted the need for more advanced and reliable security systems. Single-layer security mechanisms are no longer effective against modern threats such as hacking, identity theft, and unauthorized access. The motivation behind this project is to develop a system that overcomes the limitations of traditional security methods by incorporating multiple layers of authentication and cybersecurity features. The aim is to create a system that not only verifies user identity more accurately but also ensures secure communication and data protection. Additionally, the rise of IOT-based smart systems has created a need for integrated security solutions that are both intelligent and user-friendly. This project addresses these challenges by combining hardware-based security with software-driven cybersecurity mechanisms.

1.6. Objectives

- To design and implement a multi-factor authentication system using multiple verification methods such as password, biometric, and alert messages.
- To develop a smart security system using embedded hardware components like sensors and microcontrollers.
- To integrate cybersecurity features such as data encryption and secure communication protocols.
- To provide real-time monitoring and alert mechanisms for unauthorized access attempts.
- To enable remote access and control through a mobile or web-based interface.
- To improve overall system reliability, efficiency, and scalability for real-world applications.

Chapter – 2

Literature Review

2.1. Introduction

With the increasing need for advanced security solutions, significant developments have been made in the field of smart security systems and authentication technologies. Traditional access control methods have evolved into more sophisticated systems incorporating multi-factor authentication, which combines multiple verification techniques such as passwords, biometrics, and one-time passwords to enhance security. In addition, the integration of IOT has enabled real-time monitoring and remote access, making security systems more efficient and user-friendly. However, the growing connectivity of these systems has also introduced cybersecurity challenges, including data breaches and unauthorized network access. Therefore, recent research focuses on combining multi-factor authentication with cybersecurity measures such as encryption and secure communication protocols to develop more robust and reliable security systems.

2.2. Review of existing systems

Traditional security systems primarily rely on single-factor authentication methods such as passwords, PINs, or physical keys. While these systems are simple and cost-effective, they are highly vulnerable to unauthorized access due to password theft, duplication of keys, and guessing attacks.

To improve security, biometric-based systems such as fingerprint recognition and facial recognition have been introduced. These systems use unique human characteristics for authentication, providing better security than traditional methods. However, biometric systems alone may still face issues such as false acceptance, spoofing, and hardware limitations.

With the advancement of Internet of Things (IOT), smart security systems have been developed that integrate sensors, microcontrollers, and communication modules to provide remote monitoring and control. These systems allow users to access and monitor security features through mobile or web applications. However, many of these systems lack proper cybersecurity measures, making them susceptible to hacking and data breaches.[1]

2.3. Limitations of existing research

Despite various advancements, existing security systems have several limitations. Traditional single-factor authentication systems are highly insecure and prone to unauthorized access. Biometric systems, although more secure, can be affected by environmental conditions and may be vulnerable to spoofing attacks.

Many IOT-based smart security systems lack proper cybersecurity integration, making them susceptible to hacking, data breaches, and unauthorized remote access. Additionally, systems that rely solely on network-based authentication may fail in case of network issues or delays.

Another major limitation is the lack of integration between physical security and cybersecurity. Most existing systems focus on either access control or data protection, but not both simultaneously. This creates potential security gaps that can be exploited by attackers.[1]

2.4. Research gap

- Most existing systems rely on single or limited authentication methods, which reduces overall security and increases vulnerability to attacks.
- There is a lack of proper integration of multiple authentication factors (password, biometric) into a single unified system, leading to inefficiency and security loopholes.
- Many IOT-based security systems focus mainly on automation and remote access, but fail to implement strong cybersecurity measures such as encryption and secure communication protocols.
- Existing systems often do not provide end-to-end security, meaning physical security and data security are handled separately rather than as a combined solution.
- Limited attention is given to real-time threat detection and alert mechanisms, which reduces the system's ability to respond quickly to unauthorized access attempts.
- Many solutions depend heavily on continuous network availability, making them unreliable in case of network failure or delays.
- There is a lack of scalable and cost-effective solutions that can be easily implemented in small-scale environments such as homes and offices.
- Existing research does not sufficiently address user-friendly design and ease of use, which is important for practical adoption of security systems.

These gaps highlight the need for a comprehensive system that integrates multi-factor authentication with robust cybersecurity features, real-time monitoring, and efficient system design, which is addressed in the proposed project.

2.5. Summary

Title	Author	Inference	Literature gap
Smart IOT based Integrated Home Security System-2020	Malti Bansal, Navodit Adarsh, Nitin Kumar, Monika Meena.	- This system integrates IOT technologies to provide remote monitoring and control of home security through connected devices. - It improves real-time alerting and automation but mainly focuses on basic authentication methods like keypad and RFID.	Keypad and RFID reader
Smart Home Security System using IOT and ESP8266-2023	Pavan Kumar Sattaru(2023), Rajasree Channagiri, Khasim vali Burugula.	- The system utilizes the ESP8266 module for wireless communication, enabling efficient and low-cost smart home security with real-time notifications. - It enhances accessibility but lacks strong multi-factor authentication mechanisms for improved security.	Multifactor access
IOT-Integrated Door Sensor Solution for Enhancing Smart Home Security-2024	Ishwari Kakade, Renuka Wadetwar, Kritika Nimje.	- This work focuses on door sensor-based intrusion detection integrated with IOT for continuous monitoring and alert systems. - It increases reliability in detecting unauthorized access but has limited support for centralized web-based control platforms.	Web based platform

Table.2.1.Literature review

This chapter reviewed various existing security systems and research works related to authentication and smart security technologies. It highlighted the advantages and limitations of traditional, biometric, alert message, and IOT-based systems. The analysis revealed significant gaps, particularly in the integration of multi-factor authentication with cybersecurity measures. Based on these observations, the proposed system is designed to overcome these limitations by providing a secure, efficient, and intelligent security solution.

Chapter – 3

System Methodology

3.1. Introduction

This chapter describes the methodology adopted for the design and implementation of the Multi-Factor Smart Security System with cyber security Integration. It explains the overall system architecture, working principle, and the step-by-step process involved in user authentication and system operation. The proposed system combines both hardware and software components to provide a secure and efficient access control mechanism.

3.2. Overall System Architecture

The proposed Multi-Factor Smart Security System is designed using an embedded architecture centered around the ESP32 micro-controller. The system integrates multiple hardware modules such as a fingerprint sensor, keypad, ESP32-CAM, LCD display, and a servo motor, along with a communication interface for web-based monitoring. The power supply unit consists of a 12V adapter. The ESP32 acts as the main controller, handling authentication, processing inputs, and controlling outputs. The system supports Wi-Fi connectivity, enabling communication with a web-based platform for remote monitoring and alerts.

3.2.1. Block diagram

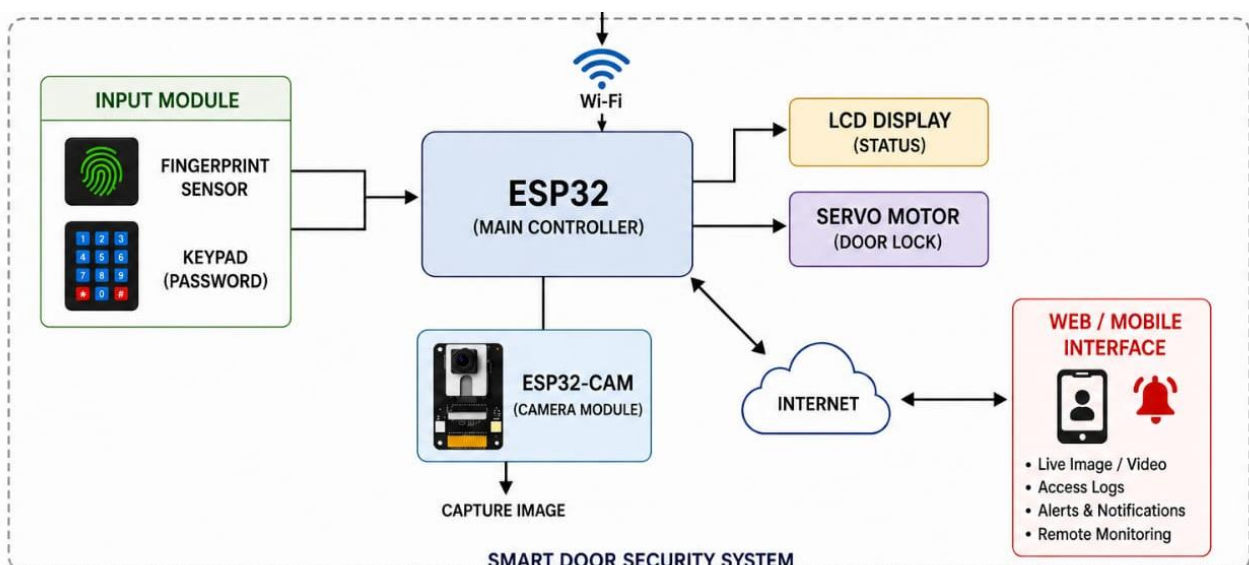


Fig.3.2.1. Block diagram

3.2.2. Block Components

3.2.2.1 ESP-32

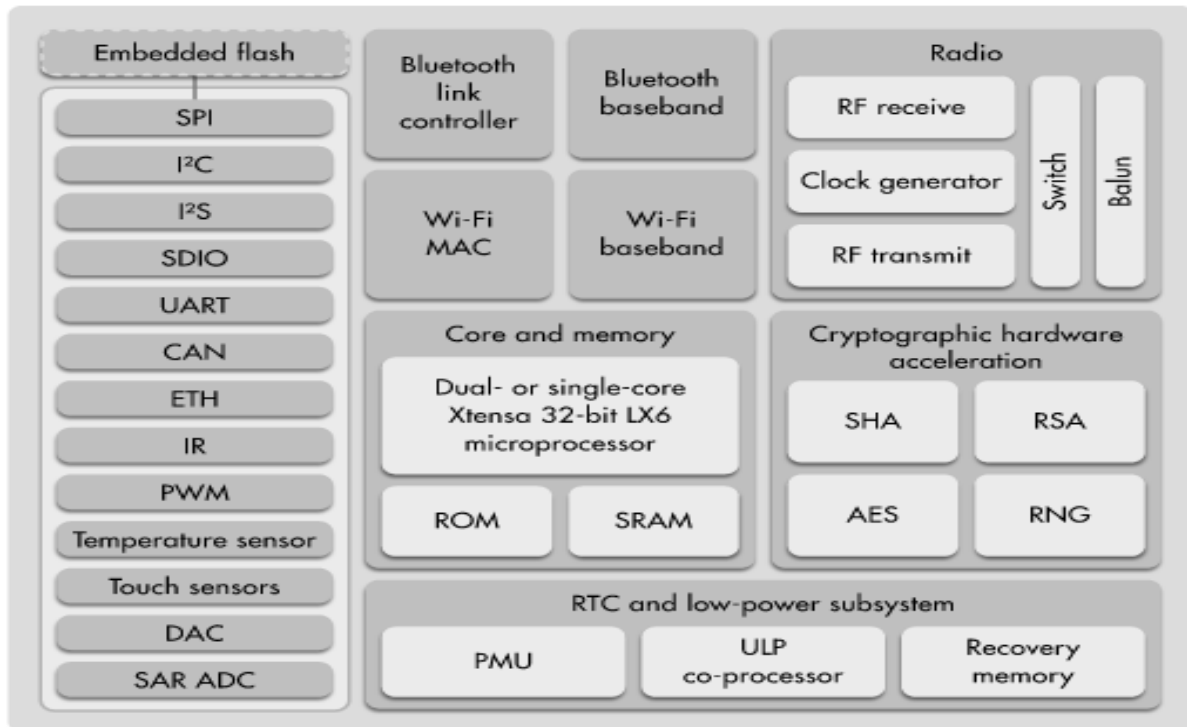


Fig.3.2.3.ESP-32 Architecture

The ESP32 is a highly integrated SoC used in IoT applications. At its core, it contains a dual- or single-core Xtensa 32-bit LX6 microprocessor along with ROM and SRAM, which handle program execution and data storage. It supports multiple communication interfaces like SPI, I²C, I²S, UART, CAN, and Ethernet, allowing it to connect easily with sensors and external devices. The chip has built-in Wi-Fi and Bluetooth modules, consisting of MAC, baseband, and RF sections (receiver, transmitter, clock generator, and antenna switch), enabling wireless communication. It also includes cryptographic hardware acceleration such as AES, RSA, SHA, and RNG for secure data processing. For analog and sensing tasks, it provides peripherals like ADC, DAC, temperature sensor, touch sensors, and PWM. Additionally, the RTC and low-power subsystem with PMU and an ultra-low-power (ULP) co-processor helps in power saving during sleep modes. Overall, this architecture makes the ESP32 a powerful, compact, and energy-efficient controller for embedded and IoT systems.

- The ESP32 is a low-cost, low-power microcontroller with integrated Wi-Fi and Bluetooth.
- Recommended operating voltage: 3.3 V, Input voltage 5 V.

- Wi-Fi active: 160 mA -260 mA, Bluetooth active 100 mA, CPU running (normal): 80 mA -120 mA.
- Up to 34 GPIO pins, Maximum current per GPIO pin: ~40 mA (absolute max), Recommended current per GPIO: ≤ 12 mA, Clock frequency: Up to 240 MHz.
- ADC resolution: 12-bit
- DAC Resolution: 8-bit.
- ESP32 supports multiple communication protocols:
 - UART (Serial communication)
 - SPI (High-speed communication)
 - I2C (Sensor interfacing)
 - I2S (Audio communication)
 - CAN (Controller Area Network – limited variants)
- Wi-Fi Standard: 802.11 b/g/n, Frequency: 2.4 GHz.
- Operating Temperature Typical range -40°C to +125°C .



Fig.3.2.2.ESP-32

The ESP32 works as an integrated system-on-chip designed for embedded and IoT applications. At its core, a dual-core Xtensa LX6 processor executes the program stored in flash memory, using SRAM for fast data access. When powered on, it initializes peripherals and begins running the user code. The ESP32 can read inputs from sensors through ADC, touch pins, or digital interfaces like I²C and SPI, process that data in the CPU, and then control outputs such as motors or LEDs using GPIO and PWM. For communication, it uses built-in Wi-Fi or Bluetooth modules to send and receive data over networks. The internal RF section handles wireless transmission and reception, while cryptographic hardware ensures secure communication. It also includes a low-power subsystem with a ULP co-processor that keeps basic functions running during sleep modes to save energy. Overall, it senses, processes, communicates, and controls—all within a single chip.

3.2.2.2 Finger print sensor

- Sensor Type: Optical fingerprint sensor
- Interface: UART (Serial communication)

- Operating Voltage: 3.3 V – 6 V
- Working Current:
 - Typical: 50 mA – 120 mA
 - Peak (during scanning): ~150 mA
- Image and Detection Specifications
 - Resolution: 500 DPI (dots per inch)
 - Image Size: 256 × 288 pixels (typical)
- Performance Specifications
 - False Acceptance Rate : < 0.001%
 - False Rejection Rate : < 1%
 - Verification Speed: < 1 second
 - Matching Mode:
 - 1:1 (verification)
 - 1:N (identification)



Fig.3.2.4.Finger print sensor

Storage Capacity typically 355 fingerprints

- Environmental Specifications
 - Operating Temperature: –10°C to +60°C
 - Humidity Range: 20% – 80% RH (non-condensing)

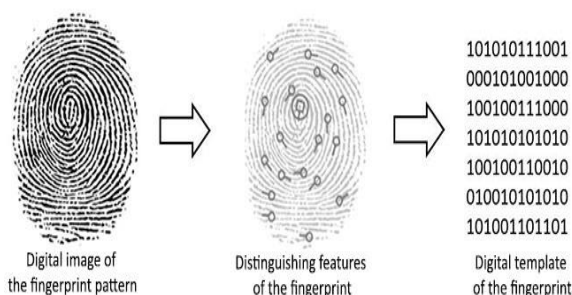


Fig.3.2.5.Finger print Reader

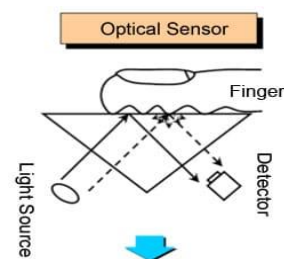


Fig.3.2.6.Optical Fingerprint sensor

An optical sensor works by shining light from a source onto the finger placed on a glass surface. Due to the principle of Total Internal Reflection, light behaves differently at the fingerprint's ridges and valleys—ridges absorb or scatter light while valleys reflect it more effectively. A detector captures this pattern of reflected light, producing a bright and dark image that represents the fingerprint. In contrast, a capacitive sensor uses an array of tiny electrodes to measure electrical changes. When a finger is placed on the sensor, the ridges come closer to the electrodes and create higher capacitance, while the valleys remain slightly farther and

produce lower capacitance. These variations in capacitance are measured and converted into a digital image of the fingerprint.

3.2.2.2 LCD Display

- Display type: alphanumeric LCD
- common model: 16x2 LCD Display
- display format: 16 characters $\times$ 2 line
- Electrical characteristics
 - operating voltage: 4.7 v to 5.3 v
 - typical voltage: 5 v
 - operating current: 1 mA to 2 mA (without backlight)
 - with backlight: 15 mA to 25 mA
- Pin configuration
 - vss: ground
 - vcc: power supply
 - vo: contrast adjustment
 - rs: register select
 - rw: read/write
 - e: enable
 - data pins: d0 to d7
 - led+: backlight positive
 - led-: backlight negative

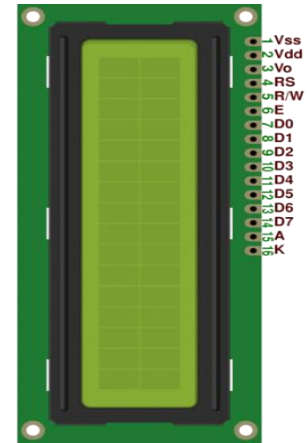


Fig.3.2.7.LCD Display

An LCD works based on the property of liquid crystals to control light. In modules like the 16x2 LCD Display, a backlight provides a constant light source, and the liquid crystal layer is placed between two polarizing filters. When no voltage is applied, the crystals allow light to pass through, making the screen appear bright. When an electric voltage is applied, the orientation of the crystals changes, blocking light and forming dark characters on the screen. The display is controlled by a driver controller such as the HD44780, which receives commands and data from a microcontroller like ESP32 or Arduino. The microcontroller sends ASCII data and instructions (like cursor position or clear display) through data pins or via I2C interface. The LCD then maps this data into characters using its internal character generator and displays them in a matrix format (e.g., 16 columns $\times$ 2 rows). By continuously updating the voltage across specific segments, the LCD forms letters, numbers, and symbols that we see on the screen.

3.2.2.3 Keypad

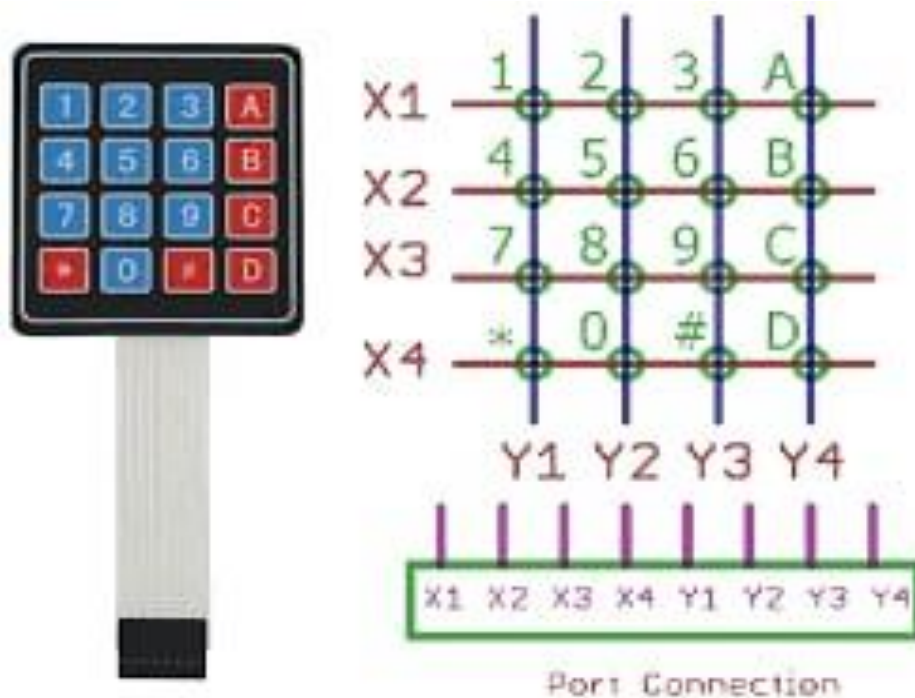


Fig.3.2.8.Keypad

A 4×4 matrix keypad is a simple input device made up of 16 push buttons arranged in a grid of four rows and four columns. Each key is positioned at the intersection of a row line (X1–X4 in your diagram) and a column line (Y1–Y4). Internally, every button acts like a switch that connects its corresponding row and column when pressed. For example, pressing the “5” key connects row X2 to column Y2, while pressing “A” connects X1 to Y4. Instead of requiring 16 separate input pins, this matrix arrangement reduces the number of connections to just 8 lines, making it efficient for microcontrollers.

The keypad works using a method called scanning. In this process, the microcontroller actively checks which key is pressed by controlling columns and reading rows. Typically, the columns are configured as outputs and the rows as inputs with pull-up resistors. The controller sets one column LOW at a time while keeping the others HIGH, then reads the row inputs. If a key in that column is pressed, the corresponding row line will also read LOW, allowing the system to determine exactly which key is pressed by matching the row and column intersection. This scanning cycle repeats rapidly, so the detection appears instantaneous to the user.

Rows	Coloumn	Key	Rows	Coloumn	Key
X1	Y1	1	X3	Y1	7
X1	Y2	2	X3	Y2	8
X1	Y3	3	X3	Y3	9
X1	Y4	A	X3	Y4	C
X2	Y1	4	X4	Y1	*
X2	Y2	5	X4	Y2	0
X2	Y3	6	X4	Y3	#
X2	Y4	B	X4	Y4	D

Table.3.2.2.1.Logic Table of keypad

3.2.2.4 ESP-32 Cam

- Type: Wi-Fi Enabled Camera Module
- Model: ESP32-CAM
- Camera sensor: OV2640 Camera Sensor
- Resolution: Up to 1600 × 1200 (UXGA)
- Wi-Fi: Built-in 802.11 b/g/n
- Electrical Characteristics Operating voltage: 3.3 V to 5 V, typical voltage: 5 V,
- operating current:

Normal mode: 80 mA to 150 mA

Wi-Fi transmission: up to 300 mA

- Integrated camera + Wi-Fi module. Supports image capture and video streaming. Built-in microSD card slot for storage. Supports face detection and recognition. Low cost and compact design. Suitable for IoT and embedded applications
- VCC: Power supply (5 V)
- GND: Ground
- U0R (RX): UART receive
- U0T (TX): UART transmit
- GPIO 0: Programming mode selection
- GPIO 2, 4, 12, 13, 14, 15: General purpose I/O pins
- 3V3: 3.3 V output
- GPIO 16: Used for camera control



Fig.3.2.9.ESP-32 Cam

The ESP32-CAM works by combining image capture, processing, and wireless communication in a single system. When powered on, the onboard ESP32 initializes its Wi-Fi module and configures the camera interface connected to the OV2640. The camera captures light through its lens and converts it into digital image data using an internal image sensor. This raw data is then processed by the ESP32, which can compress it into formats like JPEG to reduce size.

Once processed, the data can be handled in different ways depending on the program: it can be streamed live over Wi-Fi as a web server, stored in the microSD card, or sent to a cloud/server for further analysis. The ESP32 can also run basic image processing tasks like face detection or motion detection using embedded libraries. Input/output pins allow it to interact with external devices—for example, turning on an LED when motion is detected. Overall, the working involves capturing an image, processing it inside the microcontroller, and transmitting or storing it, all controlled through embedded code.

3.2.2.5 Servomotor

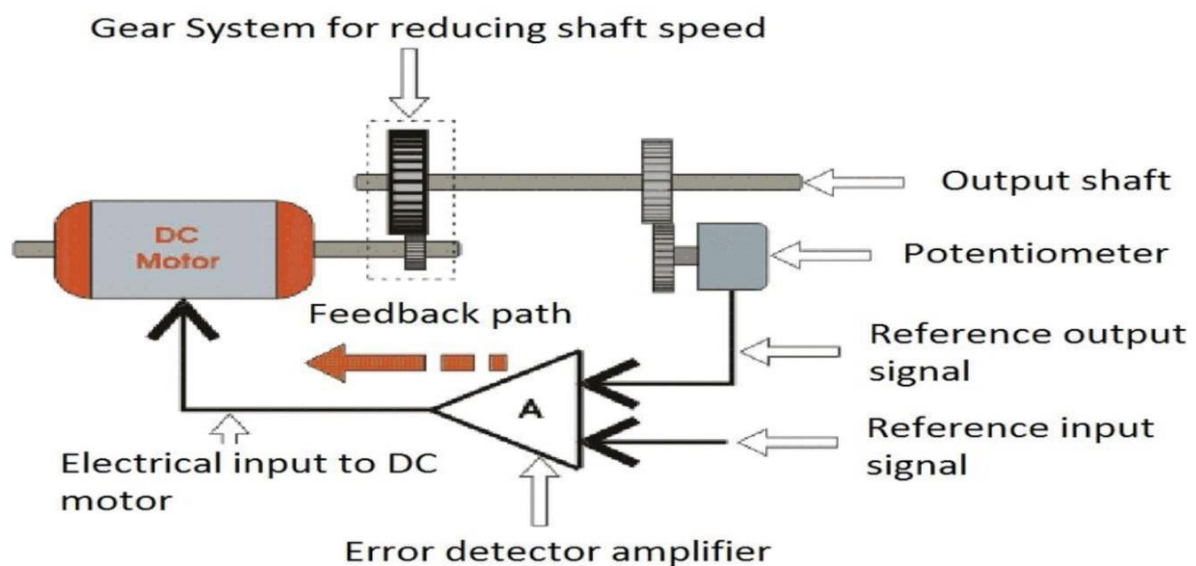


Fig.3.2.10.DC Servo motor

This diagram represents a closed-loop DC motor control system used to control shaft position or speed. When a reference input signal is given (desired position/speed), it is compared with the feedback signal obtained from the potentiometer, which is mechanically connected to the output shaft.

Chapter – 4

System Implementation

4.1. Hardware Implementation

The hardware components of the system are assembled and interconnected according to the design. The ESP32 acts as the main controller, interfacing with the fingerprint sensor, keypad, ESP32-CAM, LCD display, and servo motor. The fingerprint sensor is connected to the ESP32 for biometric authentication.

4.2 Flowchart of the system

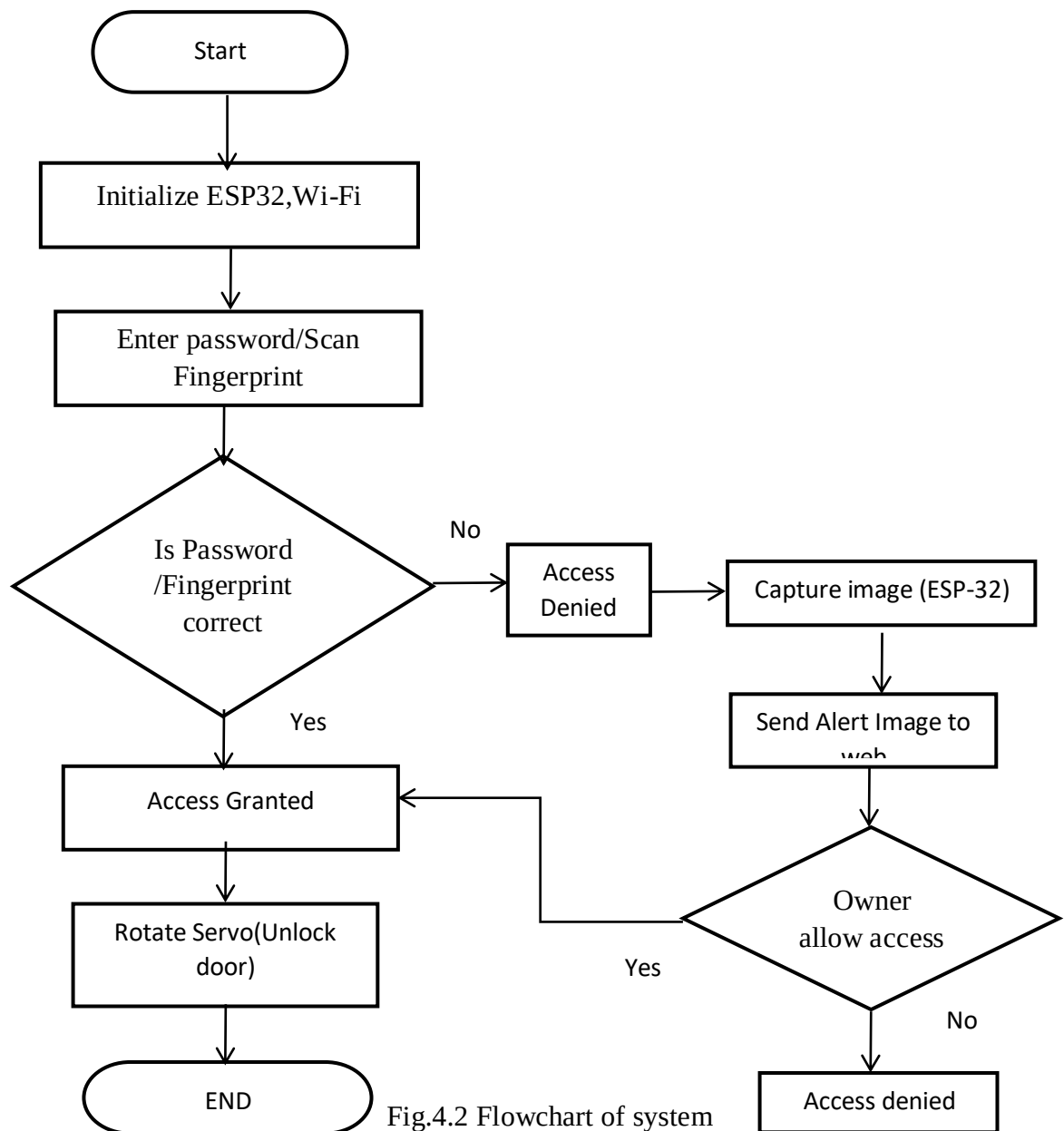


Fig.4.2 Flowchart of system

4.2.1. Functions of software

- input from keypad and fingerprint sensor
- Verifying user credentials
- Capturing images using ESP32-CAM
- Reading Sending alerts and images to the web-based platform
- Receiving user response (allow/deny)
- Controlling the servo motor for door access
- The web-based platform is developed to display alerts, images, and allow the user to control access remotely.

4.3. Hardware setup



Fig.4.3.Hardware setup

- The hardware components of the system are assembled and interconnected according to the design. The ESP32 acts as the main controller, interfacing with the fingerprint sensor, keypad, ESP32-CAM, LCD display, and servo motor.
- The fingerprint sensor is connected to the ESP32 for biometric authentication.
- The keypad is used for password entry and is interfaced through GPIO pins.
- The ESP32-CAM captures images and sends them via Wi-Fi.
- The servo motor controls the door lock mechanism.
- The LCD display shows system status messages.
- The system is powered using a 12V adapter.
- All components are tested individually and then integrated to ensure proper functioning.

SR. NO.	COMPONENT	IMAGE	SPECIFICATIONS / DESCRIPTION
1	ESP32 Microcontroller		32-bit dual-core processor - Operating Voltage: 3.3V - Wi-Fi + Bluetooth support - Multiple GPIO pins
2	Fingerprint Sensor (R305)		- Operating Voltage: 3.3V – 5V - Fingerprint storage capacity: 1000 - High accuracy and fast recognition
3	4x4 Keypad		- Matrix: 4 Rows x 4 Columns - Operating Voltage: 3.3V – 5V - Used for password input
4	I2C LCD Display (16x2)		- Display Size: 16x2 Characters - Operating Voltage: 5V - I2C Interface (SDA, SCL)
5	ESP32-CAM Camera Module		- Camera: OV2640 (2MP) - Operating Voltage: 5V - Supports image capture and streaming
6	Servo Motor (SG90)		- Operating Voltage: 4.8V – 6V - Rotation Angle: 0° – 180° - Used for door lock mechanism
7	Power Supply		- Input: 100 – 240V AC, 50/60Hz - Output: 5V DC (suitable for ESP32 & modules) - Used to power the entire system

Table.4.1. Hardware Specifications

4.4. Testing and Validations

The system is tested under different conditions to evaluate its performance:

- Authentication Testing: Verified correct and incorrect password/fingerprint inputs
- Camera Testing: Ensured proper image capture and transmission
- Alert System Testing: Checked real-time alert delivery to web platform
- Access Control Testing: Verified door unlocking and locking mechanism
- Network Testing: Tested system performance under stable and unstable internet conditions



Fig.4.4.1.Password entry using keypad



Fig.4.4.2.Access granted by entering correct password

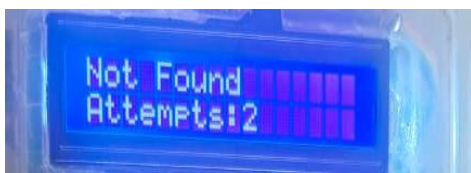


Fig.4.4.3.attempts given for wrong entries



Fig.4.4.4.If more than 3 incorrect attempts system lock for 10sec

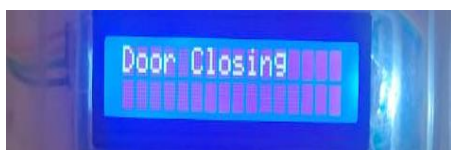


Fig.4.4.5. After person enters door closes

Chapter – 5

Results and Observations

5.1. Results

The implemented system successfully demonstrates multi-factor authentication with real-time monitoring and alert-based access control. The ESP32 processes authentication efficiently, and the ESP32-CAM captures clear images of users attempting access.

The alert system effectively sends notifications to the web platform, allowing the owner to make decisions remotely. The servo motor responds correctly based on the owner's decision, ensuring secure access control. The system provides a balance between security, usability, and real-time responsiveness.

5.2. Observations

- ✓ The system successfully performed multi-factor authentication using both password and fingerprint verification.
- ✓ The ESP32 processed inputs efficiently and controlled all components without delay.
- ✓ The camera module captured images clearly during each access attempt.
- ✓ Real-time alerts were successfully sent to the web-based platform for user monitoring.
- ✓ The owner was able to remotely approve or deny access, ensuring better control over the system.
- ✓ The servo motor responded accurately by locking and unlocking the door based on the decision.
- ✓ The system maintained reliable performance under normal operating conditions.

Chapter – 6

Discussions

6.1. Performance Analysis

The performance of the proposed Multi-Factor Smart Security System is evaluated based on parameters such as authentication accuracy, system reliability, response time, and overall security strength. By integrating multiple authentication methods, the system significantly reduces the probability of unauthorized access compared to single-factor systems. The response time of the system during user verification and alert generation is observed to be quick and efficient, ensuring real-time operation. The system also maintains consistent performance under normal conditions, providing stable and reliable functionality.

In addition, the implementation of cybersecurity features such as data encryption and secure communication enhances the protection of sensitive information during transmission. The system is capable of detecting unauthorized attempts and generating alerts promptly, which improves its effectiveness in real-world applications. Overall, the system demonstrates a balanced performance in terms of security, efficiency, and usability.

6.2. Advantages of the Proposed System

The proposed system offers several advantages, which make it more effective than traditional security systems:

- Provides enhanced security through multi-factor authentication (password, biometric, OTP).
- Reduces the risk of unauthorized access and security breaches.
- Integrates cybersecurity features such as encryption and secure data transmission.
- Supports real-time monitoring and instant alert generation.
- Enables remote access and control through mobile or web interface.
- Ensures better reliability and accuracy in user authentication.
- Can be used in multiple applications such as homes, offices, and secure areas.
- Offers scalability, allowing future upgrades and additional features.
- Provides a user-friendly interface for easy operation and management.

6.3. Limitations of the System

Despite its advantages, the system has certain limitations. The cost of implementation is relatively higher compared to traditional systems due to the inclusion of multiple hardware components and advanced technologies. The system may depend on network connectivity for features like OTP generation and remote monitoring, which can lead to delays or failures during poor network conditions. Biometric authentication may sometimes produce errors such as false acceptance or rejection due to environmental or sensor-related issues.

Furthermore, the system is mainly designed for small to medium-scale applications and may require additional optimization and resources for large-scale or industrial deployment. Power consumption and maintenance of multiple components can also be considered as minor challenges in long-term usage.

6.4. Comparison with Existing Systems

Aspect	Existing Systems	Proposed System
Authentication Method	Single-factor authentication	Multi-factor authentication (fingerprint, keypad, etc.)
Security Level	Less secure, more prone to attacks	More secure and robust
Reliability	Limited (standalone biometric or OTP-based)	High reliability due to multiple integrated methods
Cybersecurity Protection	Lacks proper protection	Includes encryption and secure communication protocols
Vulnerability to Attacks	Highly vulnerable to cyber attacks	Enhanced data safety and protection
Monitoring & Alerts	Limited or absent	Real-time monitoring and alert systems
Remote Access	Not available or limited	Supports remote access via web/mobile
Overall Efficiency	Basic security solution	Comprehensive and efficient security system

Chapter – 7

Conclusion and References

7.1 Conclusion

The project “*Design and Implementation of a Multi-Factor Smart Security System with Cybersecurity Integration*” successfully presents a reliable and intelligent solution to modern security challenges by combining multiple layers of authentication and real-time monitoring. Unlike traditional systems that rely on a single method, this system integrates password-based access with fingerprint verification, significantly reducing the risk of unauthorized entry and enhancing overall security. The use of the ESP32 microcontroller ensures efficient coordination between all hardware components, including the keypad, fingerprint sensor, LCD display, servo motor, and the ESP32-CAM. The system operates in real time, providing quick response and accurate authentication, which is essential for practical deployment. A major advancement in this implementation is the integration of the ESP32-CAM web platform, which enables the system to capture images during repeated failed access attempts and store them on an SD card for later review, thereby strengthening surveillance capabilities. Additionally, the system supports owner-controlled access, allowing users to monitor activities remotely and make decisions regarding entry approval, which adds an important cybersecurity dimension to physical security. The overall design ensures a balance between performance, cost-effectiveness, and user convenience, making it highly suitable for applications in homes, offices, and restricted areas. The system has been tested under various conditions and has demonstrated stable performance, fast processing, and dependable operation. Furthermore, this project contributes to broader developmental goals by aligning with SDG 9 (Industry, Innovation and Infrastructure) through the promotion of IoT-based innovation and enhancement of digital infrastructure, and SDG 11 (Sustainable Cities and Communities) by improving safety and security in modern living environments. In conclusion, the proposed system effectively integrates embedded systems, IoT technologies, and cybersecurity principles to deliver a smart, scalable, and user-friendly security solution that addresses the increasing demand for advanced protection in today’s interconnected world.

7.2. References

- Bansal, Malti, Navodit Adarsh, Nitin Kumar, and Monika Meena. "24× 7 Smart IoT based integrated home security system." In *2020 Fourth international conference on inventive systems and control (ICISC)*, pp. 477-481. IEEE, 2020.
- Kakade, Ishwari, Renuka Wadetwar, Kritika Nimje, Sagarkumar S. Badhiye, Pradnya Borkar, and Snehal Shinde. "IoT-Integrated Door Sensor Solution for Enhancing Smart Home Security." In *2024 4th International Conference on Technological Advancements in Computational Sciences (ICTACS)*, pp. 913-917. IEEE, 2024.
- Pramana, Dian, I. Putu Sumardika Eka Putra, Komang Gede Krisna Bayu Pamungkas, Dandy Pramana Hostiadi, Made Liandana, and Gede Angga Pradipta. "Smart Home System Based on IoT Concept Using Multi-Sensor Communication." In *2025 International Conference on Computer Sciences, Engineering, and Technology Innovation (ICoCSETI)*, pp. 278-283. IEEE, 2025.
- Sattaru, Pavan Kumar, Khasim Vali Burugula, Rajasree Channagiri, and S. Kavitha. "Smart home security system using IoT and ESP8266." In *2023 5th International Conference on Smart Systems and Inventive Technology (ICSSIT)*, pp. 469-474. IEEE, 2023.